Chain Complex Structure on the 4D Hypercubic Lattice: Structural Correspondence between Kihara Cube-Packing and Wilson Lattice Gauge Theory

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Main Result: Paper 7's α identity is mathematically isomorphic to standard Wilson lattice gauge theory The self-consistent identity from the companion paper (Paper 7),

$$\alpha^{-1} = 137 + \tfrac{\pi^2}{2}\alpha \quad \text{(8.7 ppb precision)}$$

is not mere numerical coincidence but is **structurally isomorphic as chain complexes** to the standard lattice gauge theory established by Wilson (1974), on the 4D Euclidean integer lattice $\mathbb{Z}^4$. The isomorphism is supported by Schläfli duality (tesseract $\leftrightarrow$ 16-cell self-duality), the hypercubic symmetry group B_4, and the discrete Stokes theorem.

Chain-complex reading of Paper 7's inside/outside decomposition $1 = 137\alpha + V_4(1)\alpha^2$ is the bulk-boundary normalisation on the chain complex. 137α is the tree-level contribution of bulk (C_4 interior, 137 cells); $V_4(1)\alpha^2$ is the self-energy correction of boundary (C_3 measure). This is structurally identical to QFT perturbative expansion (renormalisation).

Decisive departure from the Eddington trap Paper 7's α identity is **not numerology but a natural manifestation of the topological structure of the 4D lattice**. It is describable in the same mathematical language as Wilson's standard theory, dramatically improving peer-review robustness.

Chain complex on $\mathbb{Z}^4$

Standard hierarchy:

- C_0 vertices / C_1 links / C_2 plaquettes
- C_3 3-cells / C_4 4-cells (tesseracts)
- Hypercubic group B_4 (order 384) acts equivariantly

Two formalisms in correspondence

Wilson lattice gauge theory:

- Gauge field $U_\ell \in U(1)$ on links C_1
- Plaquette action S_p on C_2
- Wilson loops as gauge-invariant observables

Kihara cube-packing (Papers 7, 3):

- Unit cubes on C_4 ($N(1) = 137$ at $R = 3$)
- Volume deficit $\Delta(R) = V_4(R) - N(k) \sim R^3$
- Boundary measure on C_3

Connected by Schläfli duality D

Tesseract $\{4,3,3\}$ and 16-cell $\{3,3,4\}$ are mutually self-dual: $f_k(\text{Tess}) = f_{3-k}(\text{16-cell})$.

Λ side	Λ^* side
C_0 site	C_4 4-cell
C_1 link	C_3 3-cell
C_2 plaquette	C_2 plaquette (self-dual)
C_3 3-cell	C_1 link
C_4 4-cell	C_0 site

Both perform the same computation Discrete Stokes theorem ($\partial^2 = 0$):

- Wilson: plaquette sum $\to$ boundary Wilson loop (interior edges cancel pairwise)
- Kihara: bulk packing $\to$ boundary $\Delta(R)$ (interior cubes cancel via shared faces)

Two approaches starting from completely different motivations arrived at the same mathematics (Poincaré duality + B_4 symmetry + Stokes).

Main Theorem (Structural Correspondence)

> $\mathcal{C}_W$ (Wilson configuration) and $\mathcal{C}_K$ (Kihara configuration) are structurally isomorphic as chain complexes on the 4D Euclidean lattice, via the B_4-equivariant chain map D (Schläfli duality).

Corollary: Bidirectional Tool Borrowing

Wilson $\to$ *Kihara*:

- Wilson strong-coupling expansion $\to$ geometric ID of c_3
- Lattice Monte Carlo $\to$ numerical check of $N(k)$
- RG $\to$ systematic continuum limit

Kihara $\to$ *Wilson*:

- Lagrange–Jacobi four-square theorem $r_4(N) = 8\sigma(N)$ $\to$ closed forms for partition functions
- Inclusion-exclusion $\to$ analytical plaquette sums
- Central projection (Paper 2) $\to$ geometric origin of Wick rotation

Difference from Eddington tradition (strengthened)

- Relation to standard physics: isolated $\to$ **isomorphic to Wilson**
- Analytical tools: ad-hoc $\to$ **Wilson tools directly applicable**
- Peer-review robustness: no foundation $\to$ **standard mathematics (algebraic topology)**

Open Problems (shared with Paper 7)

- Mechanistic derivation of α's value
- Privilege of $R = 3$
- Geometric ID of $c_3 \approx 1.6 \times 10^{-3}$
- Generalisation to other gauge groups (SU(2), SU(3))
- Projection to physical (3+1)D spacetime

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What this paper does *not* claim

- First-principles derivation of α from the isomorphism
- Unique emergence of U(1) gauge symmetry
- Full deployment of categorical machinery (functors, natural transformations)

Note article (English) https://note.com/kiharanoriaki/n/n9c37c6f99a0a

Note article (Japanese) https://note.com/kiharanoriaki/n/n87df7a4977e7

Zenn article (Japanese, technical exposition) https://zenn.dev/noriaki_kihara/articles/alpha-isomorphism-lattice-gauge

Zenodo (Paper PDF, LaTeX, MD, JA/EN bilingual) https://zenodo.org/records/19881119

Related: Paper 7 (α-Identity v3) Zenodo / note EN / Zenn